

NEURAL BELLMAN OPERATORS

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We study neural policy evaluation, improvement, and independent verification in a continuous economy with costly preference adjustment. Feasible learned policies and market-deflator witnesses certify a positive value of access to adjustment under the original utility and first-exit contract. At the central state, the certified gain is at least .04094994 when the adjustment-cost coefficient is two. Constructive interval refinement encloses the no-adjustment optimum within .00905. The flexible policy is evaluated much more precisely than its current .10634 regret bound. For coupled nonconvex controls, a scalar dual branch-and-bound procedure certifies continuous actions through 128 dimensions, and a Cole–Hopf relaxation supplies absolute value lower bounds. A frozen, symmetrically tuned, multi-budget comparison retains all holdout outcomes, including a stronger independently specified feedback baseline. Complete proofs separate policy evaluation, optimality, computational cost, and the economic inequalities each calculation establishes.

KEYWORDS: Stochastic control, Neural policy iteration, Endogenous preferences, Recursive utility, Verification, Computational economics.

1. INTRODUCTION

A numerical decision rule is useful for economic analysis only when its error is small relative to the comparison being made. This requirement is distinct from a small training loss. It is also distinct from outperforming another learning algorithm. When a decision maker can change a preference state, an apparent welfare gain may reflect either the adjustment technology or an inaccurate evaluation of the policy without that technology. When termination is an economic contract, an inaccurate treatment of first exit can dominate both effects.

This paper studies Neural Bellman Operators (NBO): separated policy evaluation and feasible policy improvement, implemented by neural networks and assessed by independent verification. The motivating economy combines consumption, portfolio choice, stochastic risk aversion, and costly preference adjustment. The analysis retains its original running utility, continuous controls, correlated Brownian shocks, and first-exit liquidation covenant. Neither reflecting the state at a training boundary nor altering the running payoff is used to establish the economic results below.

The main economic result is a strictly positive, verified lower bound on the value of access to preference adjustment. At the central initial state and adjustment-cost coefficient $k = 2$, access raises optimal welfare by at least .04094994 in the model’s fixed utility units. The corresponding lower bounds are .06274788 at $k = .5$ and .00781212 at $k = 8$. These are comparisons with the *optimal* no-adjustment economy, not merely with an arbitrarily chosen no-adjustment policy. A feasible learned policy supplies the flexible economy’s lower bound. An auxiliary market-deflator witness supplies the restricted economy’s upper bound while allowing every original adaptive consumption and portfolio policy. A scalar state used to construct the witness does not change the economic opportunity set.

The computational argument has three parts. First, smooth one-sided witnesses bound the original stopped problem without assuming a smooth optimal value or regularity of the boundary under every control. Second, fitting and certification are separate operations. A polynomial fitted to a continuous-generator equation is converted to outward Bernstein coefficients and refined

until its residual enclosure reaches the requested level. Independently, Gaussian polynomial moments evaluate a trained time-control policy under the exact continuous stopping contract. Third, a bounded-source Feynman–Kac construction and exponential localization potentials turn the no-adjustment market-deflator relaxation into a verifiable economic upper bound. Weighted Taylor boxes refine its initial value enclosure from width .00568 to below .000320.

The distinction between evaluation accuracy and optimality accuracy matters. At $k = 2$ the original-payoff learned policy has a value enclosure of width 1.33×10^{-5} , but its current regret upper bound is .10634, not .0000133. This replaces the previous 8.33956 bound for a different retained policy with a substantially smaller original-model certificate; it does not establish the declared .01 full-policy target. The positive access-welfare result is valid nevertheless because it pairs the feasible flexible policy with a substantially tighter upper bound on the restricted optimum. The paper therefore establishes a specific economic inequality rather than interpreting an unresolved approximation as an optimal policy surface.

A second contribution addresses the inner optimization needed for verification. In a coupled nonconvex control family, all nonconvex coupling depends on the sum of the actions. A one-dimensional branch-and-bound calculation, together with strongly convex scalar dual subproblems, certifies the full continuous action box without a tensor action grid. Executed certificates reach gaps below 10^{-5} through 128 action dimensions. At the eight- and sixteen-dimensional learned-critic test states, interval differentiation of the actual network weights also encloses the error in the floating-point critic gradient. This is an implemented bridge between learned parameters and an action certificate, not an assumption that an optimizer’s stationary point is globally greedy.

The same nonconvex family is given an absolute accuracy axis. A pointwise running-cost relaxation and removal of control bounds admit a Cole–Hopf representation that retains the original nonquadratic terminal payoff. Orthogonal Gaussian coordinates integrate analytically, leaving a positive scalar integral with an outward enclosure. The resulting number is a lower bound for the unchanged original minimization problem. Relative performance is evaluated separately in a frozen holdout experiment: three dimensions, two training budgets, twelve seed pairs per panel, equal learning-rate search budgets, warmed initialization, randomized method order, and a third, independently specified feedback rule. The pinned author implementation of SOC-MartNet is not modified.

Classical verification, monotone approximation, and primal–dual numerical methods supply important foundations. The stopped Itô argument is not claimed as a new general verification principle. Nor is alternating neural evaluation and improvement claimed as a new algorithmic idea. The contribution is the constructive combination of original-contract verification, independent economic upper and lower bounds, computable neural-jet allowances, and a non-enumerative global action audit, with each error quantity matched to an executed calculation. Section 8 compares these objects theorem by theorem with neighboring methods. Recursive utility, sophisticated temporal selves, dynamic games, stochastic-trace identities, counterexamples, and the complete preceding manuscripts are preserved in the supplement rather than being used to enlarge the empirical claims of the present application.

2. THE ECONOMIC ENVIRONMENT

The state is $s = (u, x)$, where u denotes relative risk aversion and x denotes wealth. An admissible action is $a = (c, \theta, p)$ in the compact set

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

Controls are progressively measurable and have well-defined stopped state equations. The feedbacks and time controls actually evaluated below are bounded and satisfy this requirement. Before termination,

$$du_t = \theta_t dt + .05 dZ_t^u, \quad (1)$$

$$dx_t = [(.02 + .06p_t)x_t - c_t]dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

The horizon is $T = 1$, discounting is $\rho = .04$, and the open state domain is $D = (1.2, 2.8) \times (.5, 2)$. Write $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. For $k > 0$, running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

The policy payoff and optimal value are

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k = \sup_\pi J_k^\pi. \quad (5)$$

The no-adjustment value V_R restricts θ to zero and retains *all* admissible consumption and portfolio choices. It is independent of k . The cost experiment holds the controls, transition law, stopping rule, settlement, utility normalization, and consumption units fixed. A state-dependent transformation of utility would generally define another economy.

The early settlement fee is a liquidation covenant. There is no continuation consumption or wealth injection following exit. At the lower wealth boundary the largest admissible wealth drift is negative. At the upper boundary, $p = 0$ eliminates the wealth diffusion and permits an inward drift. Thus neither a silently viable rectangle nor a theorem requiring uniform controlled-boundary regularity can be substituted for this contract. The verification below requires only inequalities for smooth test functions on the trace at which the process is actually stopped.

For a smooth function v , put

$$\mathcal{L}^a v = \theta v_u + [(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2 x^2 v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$R_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

The mixed derivative is retained. Its sign is not inferred solely from the negative shock correlation.

PROPOSITION 1—Complete selection at a fixed jet: *For the generator in (7), a global maximizing action can be selected componentwise. Consumption is $\text{proj}_{[.05, .8]}(v_x^{-1/u})$ when $v_x > 0$ and is .8 otherwise. Adjustment is $\text{proj}_{[-.2, .2]}(v_u/k)$. Let $h_1 = .06xv_x - .0025xv_{ux}$ and $h_2 = .04x^2v_{xx}$. If $h_2 < 0$, the portfolio is $\text{proj}_{[-.5, .8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint maximizing $h_1p + h_2p^2/2$ is selected, with a fixed tie rule.*

The statement includes convex and zero-curvature portfolio objectives and nonpositive marginal wealth value. It applies to the continuous generator, not to the interpolated, killed-Euler finite-step target. It says nothing about global optimization over shared neural parameters.

3. LEARNING AND VERIFICATION ARE SEPARATE OPERATIONS

For a fixed policy, exact evaluation solves $R_\pi v = 0$ with the stopping contract. Exact improvement maximizes the Hamiltonian of that fixed evaluation. Denote these operators by $\mathcal{E}$ and $\mathcal{I}$; ideal policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. A neural implementation approximates both operators, while an independent evaluator measures the payoff of the policy that will actually be used.

In the raw NBO implementation, critic updates hold the actor fixed. Actor updates hold critic parameters fixed, but retain differentiation through the controlled successor state. A rival’s parameters, when present in a game, are also frozen during unilateral improvement. This separation prevents an actor loss from lowering itself by changing the object being evaluated. It does not make the joint network optimization convex. The ideal policy-iteration compactness and limiting arguments are retained in the supplement; no such argument is invoked to certify an observed Adam trajectory.

The inherited finite safeguard is a distinct algorithm. For $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, positive sub-stochastic P_n^a , and $\delta = e^{-\rho\Delta}$, define the policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and the full candidate improvement gap $\eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error is at most b , backward comparison yields

$$\|V_{h,n}^* - V_{h,n}^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

The state-dependent envelope and threshold-correction proof are reproduced in the supplement. A threshold can reduce the fraction of decisions changed while leaving the cost of a complete candidate audit essentially unchanged. In particular, the retained 125-action tensor audit is not the non-enumerative certifier of Section 6. Its floating-point envelope is not relabeled as an outward continuous-time bound.

3.1. A quantitative bridge from a fitted network

Let $C_\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible witnesses: *Witnesses are bounded, continuous on the stopped trace, and $C^{1,2}$ in the interior of each time slab. Localized Itô identities pass to the stopping time by uniform integrability. At slab interfaces an upper witness has nonpositive forward jumps and a lower witness has nonnegative forward jumps. The implemented policy has a well-defined stopped state equation.*

THEOREM 1—Stopped one-sided comparison: *Suppose Assumption 1 holds, $U \geq g - b_U$ and $L \leq g + b_L$ on the stopped trace, $\sup_a R_a U \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances are nonnegative. Then*

$$J_k^\pi \geq L - b_L - C_\rho(T-t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T-t)\epsilon_U, \quad (9)$$

$$0 \leq V_k - J_k^\pi \leq U - L + b_U + b_L + C_\rho(T-t)(\epsilon_U + \epsilon_L). \quad (10)$$

The inequalities do not require an attained optimizer or a smooth optimal value.

THEOREM 2—Cover-and-oracle certificate for neural output: *Let a continuous neural witness v and a feasible neural policy π satisfy Assumption 1, with stopped-trace error at most b . Let*

cells cover the full audited time–state domain and have representatives z_j . Suppose independent calculation proves, throughout cell j ,

$$|R_\pi v(z) - R_\pi v(z_j)| \leq \omega_j, \quad |\Gamma(z) - \Gamma(z_j)| \leq \nu_j, \quad \Gamma = \sup_a R_a v - R_\pi v.$$

Suppose the representative evaluation intervals imply $|R_\pi v(z_j)| \leq e_j$ and a global action oracle implies $\Gamma(z_j) \leq q_j$, including jet, inner-solve, and arithmetic allowances. Set $e = \max_j (e_j + \omega_j)$ and $q = \max_j (q_j + \nu_j)$. Then

$$0 \leq V_k - J_k^\pi \leq 2b + C_\rho(T - t)(2e + q). \quad (11)$$

In particular, certified derivative bounds M_j, N_j and cell radii h_j permit $\omega_j = M_j h_j$ and $\nu_j = N_j h_j$. Any sequence of learned outputs for which these verified allowances vanish has vanishing regret, irrespective of whether its parameters converge.

This is a quantitative a posteriori connection to network errors, not a claim that an empirical mean-square loss controls a supremum. Interval differentiation or weight-based derivative bounds can supply the cell moduli. They may be conservative, and covering a high-dimensional state domain remains costly. Section 6 executes the jet-to-action part for actual learned critics. The economic calculation instead fits an independent polynomial witness; it does not falsely claim to have certified a whole-state neural critic merely because its action audit passed at selected states.

4. CONSTRUCTIVE CERTIFICATES IN THE ORIGINAL ECONOMY

All initial numerical bounds in this section concern $(t, u, x) = (0, 2, 1.25)$. A central-state certificate is not advertised as a uniform certificate for every state in D . The policy used here is newly trained in a time-control neural subclass. The retained lattice-trained, state-feedback NBO policies, their finite corrections, and their previous continuous bounds remain separately available in the supplement.

4.1. Fitting an upper witness without importing an optimal value

Put $h = 1 - t$ and restrict the *verification domain*, not the economic domain, to $u \in [1.5, 2.6]$. Fit

$$F(h, u) = h \sum_{i=0}^{12} \sum_{j=0}^{14} \beta_{ij} T_i(2h - 1) T_j \left(2 \frac{u - 1.5}{1.1} - 1 \right).$$

The coefficients are obtained from a method-of-lines initializer and residual least squares. The optimizer’s collocation residual is not the certificate. For $w = G + F$, the consumption and portfolio maximizers are .8 and .8. Maximizing the remaining wealth term over $x \in [.5, 2]$ gives the scalar residual majorant

$$\mathcal{R}_k F = -F_h + .00125 F_{uu} - .04 F + H_k \{F_u - .04(u - 2)\} + \frac{.8^{1-u}}{1-u} \quad (12)$$

$$+ .0008(u - 2)^2 + .00195 + .00352 - .04 - .004 \log 2, \quad (13)$$

$$H_k(q) = \max_{|\theta| \leq .2} (q\theta - k\theta^2/2).$$

The source is the original utility, not a payoff manufactured to make F exact.

Power coefficients are enclosed at 70-digit interval precision and converted to Bernstein coefficients. The utility term is replaced by a polynomial with a proved uniform remainder below 2.117×10^{-9} . De Casteljau subdivision uses outward rounding at every elementary step. The positive-saturation, negative-saturation, and interior branches of H_k are all included whenever their certified q range permits them. Outward thresholds also cover a cell touching a switching surface. The same calculation proves $F/h \geq -8$.

PROPOSITION 2—Localized fitted upper witness: *Suppose $F(0, u) = 0$, $F/h \geq -8$, and $\mathcal{R}_k F \leq \epsilon$ on the verification rectangle. For $k \geq .5$, let*

$$Z(h, u) = 8h\{e^{-120(u-1.5)+42h} + e^{-160(2.6-u)+64h}\}, \quad (14)$$

$$U(t, u, x) = G(u, x) + F(1-t, u) + Z(1-t, u) + \epsilon_+ C_\rho(1-t). \quad (15)$$

Then $V_k(0, 2, 1.25) \leq U(0, 2, 1.25)$. No reflecting boundary or truncation of the original policy is introduced at $u = 1.5$ or 2.6 .

The exponential terms bound continuation when an artificial verification face is reached. The globally valid upper witness G is used beyond that face. At the central state the localization addition is less than 1.219×10^{-7} , rather than an eight-unit initial gap. Uniform residual correction and localization are different terms and are reported separately.

4.2. A learned feasible policy and independent stopped evaluation

The actor has one time input, a twelve-unit tanh layer, and two outputs. At sixteen fixed time nodes, a sigmoid constrains adjustment to $[0, .2]$, and a differentiable budget projection constrains consumption to $[.05, .8]$. The portfolio is zero. Training uses the original expected payoff, Gaussian quadrature, and no optimal-value or greedy-action labels. The deployed controls are the resulting frozen binary64 constants on each time slab; there is no claim that this time-control class exhausts the state-feedback opportunities in (5).

The budget projection enforces

$$\sum_{j=0}^{15} c_j \int_{j/16}^{(j+1)/16} e^{-.02r} dr < 1.25 - .5e^{-.02}. \quad (16)$$

Thus $x_t = e^{.02t} \{1.25 - \int_0^t e^{-.02r} c_r dr\}$ is deterministic. Since consumption exceeds $.02(1.25)$, wealth decreases and its interval-enclosed terminal value remains strictly above $.5$. Wealth never exits. The remaining preference exit is controlled by a Brownian maximal inequality, not by a discretized boundary check.

An evaluator independent of the training quadrature approximates utility by a degree-90 polynomial in $u - 2$, with a uniform analytic remainder on the original preference interval. Free Gaussian moments are evaluated by recurrence, integrated over outward time rectangles, and corrected for the stopped event using a maximal probability bound and Cauchy–Schwarz. The terminal wealth, adjustment expenditure, polynomial approximation, time integration, exact-exit correction, and arithmetic are all enclosed. There is no Euler, action-grid, or spatial-interpolation term in this route. Those omissions are consequences of the deployed policy and analytic evaluation, not uncomputed entries assigned zero.

Table I reports the actual original-economy bounds. Table II makes the constructive refinement explicit. The last row gives substantially more precise policy evaluation than optimality

TABLE I
ORIGINAL-PAYOFF CONTINUOUS CERTIFICATES AT THE CENTRAL STATE

k	Policy payoff enclosure	Optimal-value upper	Regret upper
0.5	$[-1.27324522, -1.27322724]$	-1.16618072	0.10706450
2	$[-1.29504316, -1.29502986]$	-1.18870455	0.10633861
8	$[-1.32818098, -1.32817412]$	-1.22254633	0.10563464

Payoff endpoints and upper bounds are rounded outward. Each regret upper is the independent optimal-value upper minus the lower endpoint of the learned policy's payoff interval. All controls in the optimal-value bound are continuous. The .01 full-policy target is not met in any row.

TABLE II
EXECUTED REFINEMENT FOR THE $k = 2$ CERTIFICATE

Time boxes	Payoff width	Sec.	Bernstein boxes	Residual upper	Sec.	Regret upper
1024	0.000209	0.363	36	0.00477	0.079	0.110237
4096	5.25e-05	0.614	57	0.00195	0.116	0.107396
16384	1.33e-05	1.618	86	0.000894	0.186	0.106339

Time boxes refine the independent policy evaluator. Bernstein boxes refine the already fitted upper witness. The two seconds columns measure the respective integration and subdivision stages; complete timings, including coefficient construction, are in the raw records. A shrinking evaluation or residual allowance need not eliminate the approximation gap between the chosen upper witness and policy.

certification. The fitted witness is independent of the policy and valid over the full original wealth and action ranges. Its loss of wealth-specific information is economically consequential. A separate wealth-sensitive linear-programming pilot is retained with its computation log and rejected candidate diagnostics; an unvalidated collocation objective is not used in either table.

5. CERTIFYING THE VALUE OF PREFERENCE ADJUSTMENT

For each feasible policy write $J_k(\pi) = A(\pi) - kB(\pi)$, where

$$B(\pi) = \mathbb{E}^\pi \int_t^\tau e^{-\rho(r-t)} \theta_r^2 / 2 \, dr, \quad 0 \leq B(\pi) \leq .02C_\rho(T-t).$$

The policy set is common to every k .

THEOREM 3—Accuracy-calibrated cost comparisons: *The optimal value is convex and nonincreasing in k . If optimizers exist, their adjustment budgets are nonincreasing across distinct prices. More generally, let π_i be ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (17)$$

If $V(k) \in [\underline{V}(k), \overline{V}(k)]$, the budget of an optimizer at k , for $k_- < k < k_+$, lies in the primitive budget range intersected with

$$\left[\frac{\underline{V}(k) - \overline{V}(k_+)}{k_+ - k}, \frac{\overline{V}(k_-) - \underline{V}(k)}{k - k_-} \right]. \quad (18)$$

The theorem concerns discounted adjustment expenditures, not a pointwise ordering of controls. A re-solved V_u/k is not a comparative-static proof. The central full-value enclosures

in Table I are still too wide for informative optimal-budget secants. The tightly enclosed expenditures in Table IV are those of the specified learned policies.

5.1. A market-deflator witness for the restricted optimum

A tighter comparison becomes possible by exploiting the restricted economy's structure. Introduce the auxiliary process

$$dY_t = .02Y_t dt - .3Y_t dZ_t^x, \quad Y_0 = 1.55. \quad (19)$$

It is a test-process constructed from the original shocks. It does not restrict portfolio information, change wealth, or add a traded asset. For

$$W_0(u, x, y) = y(x - .5) + .1 \log(.5) - .02(u - 2)^2,$$

the p terms in the generator of Yx cancel exactly. With $\theta = 0$, define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad (20)$$

$$S(u, y) = \Psi(u, y) + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2. \quad (21)$$

On $u \in [1.5, 2.6]$, $y \in [.2, 12]$, one has $-8 < S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$ by clipping its two arguments at these verification faces. Let

$$F_R(h, u, y) = \mathbb{E}_{u,y} \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (22)$$

where $dU_r = .05 dZ_r^u$ and Y follows (19). Thus $-8h \leq F_R \leq 0$.

THEOREM 4—A restricted-economy upper bound: *Let Z_R be the sum of the following nonnegative localization potentials:*

$$Z_{R,u} = 8h \{ e^{-200(u-1.5)+50h} + e^{-240(2.6-u)+72h} \}, \quad (23)$$

$$Z_{R,y} = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (24)$$

Then

$$V_R(0, 2, 1.25) \leq W_0(2, 1.25, 1.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (25)$$

The upper bound covers all adaptive original consumption and portfolio controls with $\theta = 0$. Consequently, for any feasible flexible policy with certified payoff lower bound $\underline{J}_k$ and any upper bound $\bar{V}_R$ from (25),

$$V_k(0, 2, 1.25) - V_R(0, 2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (26)$$

The expectation in (22) has an explicit two-Gaussian representation. We transform time by $r = v^2$, evaluate constant and linear Taylor terms with interval-enclosed weighted moments, and bound the Hessian remainder on each box. Exponential Taylor remainders, Gaussian tails, and localization are included. This produces the executed refinement in Table III; it does not appeal to a mesh difference as an error estimate. The final restricted-optimum upper bound

TABLE III
REFINING THE NO-ADJUSTMENT DUAL WITNESS

Weighted boxes	Dual witness-value enclosure	Width	Seconds
16384	$[-1.33876765, -1.33309541]$	0.00567	2.67
131072	$[-1.33676857, -1.33544680]$	0.00132	15.13
1048576	$[-1.33631272, -1.33599309]$	0.00032	110.24

These intervals enclose the *dual witness value*, not the restricted optimum. Only their upper endpoints are used as upper bounds for that optimum. A separately evaluated restricted policy gives the primal lower bound. Original stopping and all adaptive consumption and portfolio opportunities are retained.

TABLE IV
CERTIFIED ORIGINAL-MODEL EXPENDITURES AND VALUE OF ACCESS

k	Policy adjustment budget	Access gain: lower	Access gain: enclosure
0.5	$[0.01960483159556, 0.01960483159557]$	0.06274788	$[0.06274788, 0.17885870]$
2	$[0.01242015505497, 0.01242015505498]$	0.04094994	$[0.04094994, 0.15633486]$
8	$[0.00192618262027, 0.00192618262028]$	0.00781212	$[0.00781212, 0.12249308]$

The expenditure column evaluates the displayed learned policy. The access columns bound the difference between the two *optimal* values. The lower endpoint pairs a feasible flexible policy with the restricted dual upper; the upper endpoint pairs the flexible upper with a feasible restricted policy. The positive lower endpoints are informative economic guarantees even though the two-sided access intervals are not narrow estimates of the welfare magnitude.

is -1.33599309 after outward display rounding. The evaluated restricted policy has payoff in $[-1.34503942, -1.34503585]$, so the restricted problem itself has an initial optimal-value enclosure narrower than .00905.

The welfare lower bounds remain strictly positive at all three prices. At $k = 2$, the quadrature enclosure of the restricted witness is more than two orders of magnitude narrower than the certified access gain. This comparison establishes a minimum economically meaningful gain without assuming that the flexible learned policy is optimal. It does not identify the exact access gain, an optimal adjustment-budget magnitude, or an optimal pointwise policy ordering.

6. GLOBAL ACTION CERTIFICATION WITHOUT A TENSOR GRID

Consider the following minimization problem, which is also used for the external comparison:

$$dX_t = 2a_t dt + \sqrt{.32} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \quad (27)$$

$$f_d(a) = \frac{1}{d} \sum_{i=1}^d \{a_i^2 + .1 \sin^2 a_i\} + .3 \sin \left(\sum_i a_i \right), \quad (28)$$

$$q_d(x) = \frac{1}{d} \sum_{i=1}^d (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right), \quad (29)$$

$$V_d^* = \inf_a \mathbb{E} \left[\int_0^1 f_d(a_t) dt + q_d(X_1) \right].$$

The objective has genuinely coupled nonconvex actions. No analytic optimal-action or optimal-value labels are supplied to training. At a fixed critic gradient p , the action-dependent Hamilto-

TABLE V
EXECUTED CONTINUOUS-ACTION CERTIFICATES

Action dim.	Tolerance	Verified gap	Scalar boxes	Convex subproblems	Seconds
8	1e-03	0.000258	33	264	0.299
8	1e-05	6.72e-06	47	376	0.493
16	1e-03	0.000931	37	592	0.476
16	1e-05	4.42e-06	57	912	0.751
32	1e-03	0.00068	85	2720	1.498
32	1e-05	3.52e-06	103	3296	1.991
64	1e-03	0.000647	163	10432	6.041
64	1e-05	3.06e-06	185	11840	5.955
128	1e-03	0.000948	317	40576	19.229
128	1e-05	4.68e-06	339	43392	20.651

One frozen jet is used per dimension, at both tolerances. At $d = 8, 16$, these are gradients of retained learned critics at recorded states; displayed gaps include interval neural-derivative allowances. At larger dimensions they are independently generated jets for the same Hamiltonian family. The experiment measures certificate work for these instances, not an average-case scaling law or a whole-state neural PDE residual.

nian is

$$H_p(a) = \sum_i f_i(a_i) + .3 \sin z, \quad f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

Each f_i is strongly convex with modulus at least $m = 1.8/d$.

THEOREM 5—Scalar global partition and convex dual bounds: *For any multiplier λ and $z \in [-d, d]$, put*

$$D(\lambda) = \sum_i \min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

On a scalar interval $z \in [z_0 - r, z_0 + r]$, a lower bound for the full Hamiltonian is

$$D(\lambda) + \lambda z_0 + .3 \sin z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \quad (30)$$

Any feasible action supplies an upper bound. Each inner convex minimum can itself be bounded below by $q(s_0) - e(s_0)^2/(2m)$, where $q = f_i - \lambda s$ and e is the derivative magnitude in a feasible descent direction, or the full derivative magnitude at an interior point. Outward versions of these bounds, followed by scalar subdivision and pruning, give a global gap on $[-1, 1]^d$.

If a stored gradient $\hat{p}$ differs from the true neural gradient by componentwise bounds δ_i , the feasible action's true-gradient gap is at most its stored-gradient gap plus $4 \sum_i \delta_i$.

Floating-point root finding only proposes multipliers and actions. It is not trusted to solve the convex subproblems exactly. Even an inaccurate proposal remains valid after the strong-convexity allowance. This structure removes tensor enumeration; it does not establish a dimension-free complexity theorem or solve a generic nonseparable nonconvex Hamiltonian.

The 128-dimensional 10^{-5} calculation uses 339 scalar boxes and 43,392 scalar convex subproblems. It replaces neither raw actor training nor the preceding finite tensor safeguard in retrospect. At the learned eight- and sixteen-dimensional jets, interval forward differentiation of the stored critic weights bounds the ℓ^1 gradient discrepancies below 4.584×10^{-17} and

TABLE VI
ABSOLUTE LOWER BOUNDS FOR THE ORIGINAL NONCONVEX CONTROL PROBLEM

d	Quadrature cells	Relaxed-value enclosure	Original-value lower	Seconds
8	1024	$[-0.08599082, -0.08384626]$	-0.08599082	0.260
8	4096	$[-0.08518701, -0.08465086]$	-0.08518701	1.081
8	16384	$[-0.08498598, -0.08485194]$	-0.08498598	4.506
16	1024	$[-0.20520651, -0.20360335]$	-0.20520651	0.271
16	4096	$[-0.20460665, -0.20420585]$	-0.20460665	1.067
16	16384	$[-0.20445644, -0.20435623]$	-0.20445644	4.212
32	1024	$[-0.28757809, -0.28684340]$	-0.28757809	0.245
32	4096	$[-0.28730394, -0.28712026]$	-0.28730394	0.988
32	16384	$[-0.28723515, -0.28718923]$	-0.28723515	4.172

A row encloses the relaxed value. Its lower endpoint is also a lower bound for the original bounded-control value. The width shown by successive enclosures is quadrature uncertainty; it does not include or estimate the structural relaxation gap.

7.877×10^{-17} , respectively. These allowances are propagated into the action gap. Theorem 2 identifies the additional state-cover and residual quantities that would be needed for a whole-policy guarantee.

7. AN ABSOLUTE BOUND AND A FROZEN EXTERNAL COMPARISON

7.1. A rigorous lower bound for the unchanged nonconvex family

Let $r_0 = 1 + .1 \sin^2(1)$, $\sigma^2 = .32$, and $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1)a^2$ for $|a| \leq 1$, replacing the running cost by $r_0 \|a\|^2 / d - .3$ and allowing unbounded controls gives a relaxation. The original terminal function q_d is retained.

PROPOSITION 3—Scalar-integral absolute lower bound: *Let Z be normal with mean $2\sqrt{d}/(r_0 + 4)$ and variance $\sigma^2 r_0 / (r_0 + 4)$. Then*

$$V_d^* \geq \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \quad (31)$$

An outward enclosure of the positive expectation yields a rigorous numerical lower bound for the original objective (29).

The relaxation is an upper/lower-bound device, not a new training problem with a known manufactured value. The terminal cosine, which couples all state coordinates, remains in the scalar expectation. Its tails are bounded analytically and its interior is integrated by interval rectangles.

7.2. Holdout design and exact deployment interpretation

The comparison retains the original actor and critic architectures: two tanh hidden layers with 48 units each, a bounded actor output, and the same terminal-value embedding. NBO uses separated evaluation and improvement with Adam, three critic steps per actor step, 256 training states, and four successors. SOC-MartNet uses its pinned author solver, RMSprop, the 600-output test network, and the inherited $J = 2$, $K = 1$ multiplier setup. These different optimization systems are disclosed rather than described as identical implementations.

The committed protocol freezes dimensions 8, 16, 32 and training budgets of 10 and 30 seconds. At each dimension, both methods receive the same six validation trials: two seeds

TABLE VII
FROZEN HOLDOUT COMPARISON: TWELVE SEED PAIRS PER PANEL

d	Budget	NBO cost	SOC cost	LQ cost	Median difference	Adjusted p
8	10	0.5812	1.0842	0.5121	-0.4783	0.001465
8	30	0.5785	1.0557	0.5121	-0.4348	0.001465
16	10	0.4663	0.7893	0.1740	-0.1672	1
16	30	0.4630	0.8456	0.1740	-0.2878	0.438
32	10	0.3813	1.2011	0.0854	-0.8691	0.001465
32	30	0.3799	1.0218	0.0854	-0.6962	0.001465

Smaller realized cost is better. The median difference is NBO minus SOC-MartNet. The last column is a one-sided exact sign-test p -value adjusted across the six predeclared dimension–budget panels. The sign test concerns the joint training-and-audit pipeline under independent seed draws, not an outward numerical error guarantee.

TABLE VIII
ABSOLUTE REFERENCE EXCESS AND ACTUAL TRAINING CLOCKS

d	Budget	Lower bound	NBO excess	SOC excess	NBO sec.	SOC sec.
8	10	-0.08499	0.6662	1.1692	10.002	10.006
8	30	-0.08499	0.6635	1.1407	30.004	30.008
16	10	-0.20446	0.6707	0.9938	10.002	10.005
16	30	-0.20446	0.6674	1.0500	30.004	30.006
32	10	-0.28724	0.6686	1.4883	10.003	10.007
32	30	-0.28724	0.6672	1.3090	30.003	30.007

Excess is the estimated mean policy cost minus the rigorous original-value lower bound. It provides an absolute reference axis but is not itself a deterministic regret enclosure: sampling error and floating-point rollout evaluation are not outward certified. Training clocks include completed-update overshoot and exclude separately recorded warmup and setup.

and three learning-rate multipliers. Selection minimizes mean independent validation cost, with a fixed tie rule. Twelve different holdout seeds, 720–731, are then evaluated at each dimension–budget pair. Within each pair, actor and critic starting weights are identical and their hashes are checked. Framework warmup precedes both clocks, and method order is seed-randomized. Training-only and setup-plus-training clocks are retained. The baseline $a_i(t, x) = \text{proj}_{[-1, 1]}[-2(x_i - .5)/\{1 + 4(1 - t)\}]$ is specified independently from the associated quadratic problem and is evaluated on the unchanged nonquadratic objective.

Each trained network is deployed as a sample-and-hold control on 128 dates and evaluated with 4,096 independent, paired Gaussian paths. In (29), drift under a held action and volatility are constant on a slab; the Gaussian transition and running-cost integral are therefore exact in real arithmetic for *that deployed continuous-time policy*. A 64-date evaluation uses aggregated common increments and changes the deployment policy. Its difference from the 128-date result is not called an Euler-error bound. Floating-point network evaluation and Monte Carlo uncertainty remain distinct from the outward deterministic lower bound in Table VI.

The holdout results favor NBO over the author implementation in mean cost in all six panels, but their interpretation is not uniform. In dimensions 8 and 32, NBO has lower paired cost for all twelve seeds at both budgets; each multiplicity-adjusted sign-test value is .001465. In dimension 16 the adjusted values are 1 and .438, so these panels do not establish a robust seed-level ordering under the declared test. Tripling the budget gives only small improvements in NBO mean cost, from .5812 to .5785, .4663 to .4630, and .3813 to .3799 respectively. Most importantly, the independently specified quadratic-feedback baseline has lower mean cost than both learned methods in every panel. Its mean costs are .5121, .1740, and .08544 in dimensions 8, 16, and 32. These retained results support the stated NBO–SOC-MartNet comparison at the

frozen budgets, not a claim that NBO is the best available control rule or that additional training has already reached an accurate optimum.

The raw records retain every seed difference, the median and interquartile range, approximate Student intervals, costs for all three rules, path standard errors, training curves, optimizer clocks, and checkpoints. Approximate Student intervals are descriptive; the exact sign calculation avoids assuming Gaussian seed-level differences. The source protocol predates holdout execution, while the benchmark family and adapters themselves were developed in earlier revisions. Holdout seeds do not turn one problem family into evidence of general superiority across economic environments. Two budget points are not a matched-accuracy frontier, and a finite absolute lower bound need not be tight enough to rank methods at a small prescribed regret.

8. RELATION TO RIGOROUS CONTROL AND NEURAL COMPUTATION

The strongest claims should be compared with methods that establish bounds, not only with methods that train networks. Theorem 1 is an application of classical stopped verification to smooth upper and lower witnesses. Its role here is to handle the actual liquidation trace without presupposing a smooth or boundary-regular optimal value. Theorem 2 supplies a computation-facing cover, derivative, and oracle allowance. Neither theorem claims that classical verification was absent from the literature.

Monotone, stable, consistent approximations with comparison provide convergence to viscosity solutions in the framework of Barles and Souganidis (1991). Controlled Markov-chain approximation is developed systematically by Kushner and Dupuis (2001); adaptive sparse-grid methods address high-dimensional dynamic models in Brumm and Scheidegger (2017). Such results concern specified approximation schemes and their hypotheses. The finite telescoping budget in the preserved manuscript names the operator discrepancies needed for a continuous statement; an observed mesh difference does not supply them. The original-payoff certificates here instead evaluate continuous generators or exact Gaussian moments directly, with separately bounded analytic and arithmetic remainders. They do not assert a convergence rate for the inherited killed-Euler scheme.

Primal–dual numerical control methods also pair feasible decisions with bounds. Brown, Smith, and Sun (2010) relax nonanticipativity with penalties and establish weak and strong duality for stochastic dynamic programs. Brown and Smith (2011) use related bounds in portfolio optimization with transaction costs. Our no-adjustment construction is a market-deflator witness with a bounded source and explicit stopped localization; it is not a claim that policy-plus-upper-bound analysis is new. Lasserre, Henrion, Prieur, and Trélat (2008) develop occupation-measure and moment relaxations for polynomial nonlinear optimal control under their stated assumptions. We do not invoke their deterministic convergence theorem for the present logarithmic, stochastic, controlled-exit problem. The explicit source bounds, interval polynomial conversions, Gaussian tails, and trace inequalities are verified here instead.

Proposition 2 and Theorem 4 identify the model-specific constructions responsible for the new economic bounds. The former couples a fitted generator witness to a small localization potential; the latter cancels every portfolio term before computing a bounded-source expectation. Theorem 5 exploits rank-one action coupling and strong convexity of the remaining scalar problems. It is a structured convex-duality and branch-and-bound construction, not a generic replacement for global nonconvex optimization. Proposition 3 preserves the nonquadratic terminal payoff in a computable relaxation of the exact comparison problem.

Neural policy iteration and neural PDE solvers are important computational antecedents. Jacka and Mijatović (2015) analyze continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) separate iteration, policy-network, and residual errors in entropy-regularized

control; Kim, Kim, Kim, and Cho (2026) study interior bounds and the mismatch between collocation and closed-loop performance. Lee and Kim (2025) combine policy iteration and operator learning. Cai, Fang, and Zhou (2025) impose martingale conditions without an explicit Hamiltonian infimum. NBO’s raw training comparison is with a pinned implementation of the last method, not a locally recreated approximation described as author code. Our certificates can also audit a policy produced by another solver. This portability is a property of independent verification, while the training results remain evidence about the specifically recorded NBO and SOC-MartNet adapters.

Recursive utility, temporal-self equilibria, and games require their own evaluation operators and optimality notions. Their complete equations, assumptions, finite gain bounds, failures, and earlier numerical results are preserved in the supplement. No new large-scale recursive or game accuracy result is inferred from the additive-utility certificates above. The stochastic-trace bias identities and local-optimizer counterexamples are preserved for the same reason: they identify conditions that a neural training loss cannot replace.

9. CONCLUSION

The computations establish a positive minimum value of access to preference adjustment in the original continuous economy. They do so by combining a feasible learned policy with an independently certified upper bound on the restricted optimum, rather than by assuming either learned rule is optimal. The restricted economy is enclosed within .00905 at the central state. The flexible policy’s evaluation is much more accurate than its present .10634 regret bound at $k = 2$; the full .01 flexible-policy objective remains a distinct numerical target.

The methodological results give operational ways to improve and audit these calculations: fitted Bernstein witnesses, stopped Gaussian-moment evaluation, market-deflator localization, interval neural derivatives, and a non-enumerative continuous-action oracle. The coupled comparison problem also has a rigorous absolute lower bound and a frozen multi-budget, multi-dimensional holdout record. Together these results sharpen the connection between neural computation and economic conclusions while leaving every inherited derivation, experiment, and unresolved diagnostic available for further review.

APPENDIX A: VERIFICATION AND COMPUTATIONAL ERROR

A.1. Selection, stopped comparison, and the neural bridge

PROOF OF PROPOSITION 1: For a fixed jet the three controls enter additively. The consumption objective has derivative $c^{-u} - v_x$ and second derivative $-uc^{-u-1} < 0$. If $v_x \leq 0$ it is increasing; otherwise its unconstrained maximizer is $v_x^{-1/u}$. The adjustment objective is $\theta v_u - k\theta^2/2$, which is strictly concave for $k > 0$. The portfolio objective is $h_1p + h_2p^2/2$. A negative h_2 gives the projected vertex. A nonnegative h_2 gives a maximum at an endpoint, including the linear and constant cases. Compactness and the stated tie rule give a feasible measurable selection. These are control-space statements at a fixed jet, not parameter-space convergence statements. *Q.E.D.*

PROOF OF THEOREM 1: Localize the process on interior compact sets and apply discounted Itô’s formula to an upper witness on each time slab. The discounted terminal witness plus accumulated reward equals the initial witness, the expected integral of $R_a U$, and the expected sum of discounted forward jumps. The latter sum is nonpositive. Boundary comparison gives $g \leq U + b_U$. Nonnegativity of discounting and the deterministic bound $\tau \leq T$ therefore imply

$$J_k^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T - t)\epsilon_U.$$

Uniform integrability permits removal of localization. Taking the supremum over all admissible controls proves the optimal-value bound; attainment is unnecessary. Apply the same identity to L along the implemented policy. Its forward jumps are nonnegative, $g \geq L - b_L$, and $R_\pi L \geq -\epsilon_L$, giving the lower bound. Subtraction proves the regret statement. The process is stopped at the economic exit; there is no use of an unproved interior trace of the optimal value. *Q.E.D.*

PROOF OF THEOREM 2: The cover and representative intervals imply $|R_\pi v| \leq e$ and $0 \leq \Gamma \leq q$ everywhere. Hence $\sup_a R_a v = R_\pi v + \Gamma \leq e + q$ and $R_\pi v \geq -e$. Apply Theorem 1 with $U = L = v$, $b_U = b_L = b$, $\epsilon_U = e + q$, and $\epsilon_L = e$. A derivative bound with respect to the chosen cell norm gives the two moduli by integration along segments, or along a specified path inside the cell. In a nonconvex cell, a verified path-length radius must replace the Euclidean radius. The vanishing-allowance assertion follows directly from the bound and does not require a compact set of network parameters. *Q.E.D.*

The finite counterpart follows from positivity of the continuation operators: propagate the evaluation error with one factor δ , propagate the optimality error with one factor δ , and add the candidate gain. Backward induction from terminal error b gives (8). The full state-dependent induction, correction rule, and recursive-operator versions are reproduced in the preservation supplement. Unlike this finite identity, a continuous interpretation additionally needs either verified operator discrepancy terms or an independent continuous verification route. The present original-payoff calculation uses the latter.

A.2. Arithmetic convention

The mathematical constants specifying the economy are exact decimals. Stored network weights, policy constants, and fitted coefficients are interpreted as exact binary64 numbers. Constant formation and polynomial coefficient conversion use 70-digit interval arithmetic; scalar optimization bounds use at least 50 digits. Vectorized interval addition, subtraction, multiplication, division by a strictly positive interval, and pairwise summation round both endpoints outwards with the adjacent binary64 number. An interval square may be enclosed by ordinary interval multiplication; this is conservative even when its lower endpoint is negative. No transcendental library approximation is treated as a rigorous interval merely because its output was subsequently rounded once. Exponentials in vectorized routines are replaced by Taylor polynomials with analytic remainders; interval logarithms, trigonometric constants, and square roots are formed in the multiprecision layer.

All derivative and residual claims concern the mathematical function represented by the stored constants, not the real-time floating-point implementation of an arbitrary neural policy. The original time controls are themselves stored constants, so their continuous implementation is unambiguous. External-network rollouts are explicitly floating-point performance estimates. This distinction is retained in every numerical table.

APPENDIX B: ORIGINAL-ECONOMY FITTED WITNESSES

B.1. Global fallback and localized upper bound

For $G = -.02(u - 2)^2 + .1 \log x$, consumption and portfolio selection give $c = .8$ and $p = .8$. The adjustment gain is bounded above by $.000512/k \leq .001024$ for $k \geq .5$. The preference part of $-\rho G$ is at most $.000512$. Consumption utility is maximized at the upper endpoint $u = 2.8$

on the stated domain. The portfolio drift and diffusion contributions at $p = .8$ are .00352. Maximizing $-.08/x - .004 \log x$ over $[.5, 2]$ gives $-.04 - .004 \log 2$. Consequently

$$\sup_a R_a G \leq \frac{.8^{-1.8}}{-1.8} + .000512 + .001024 + .00195 + .00352 - .04 - .004 \log 2 < -.8659.$$

Also $G \geq g$. Stopped verification gives $V_k \leq G$ globally on the original domain, including as a continuation bound at an artificial verification face.

PROOF OF PROPOSITION 2: The derivatives of $G + F$ in x coincide with those of G . Thus the same c and p selection applies. The residual after maximizing over wealth is precisely (13). For $z = 8h \exp\{\lambda(u - u_0) + \kappa h\}$, where $h = 1 - t$,

$$(\partial_t + \mathcal{L}^a - \rho)z = 8e^{\lambda(u - u_0) + \kappa h} \{-1 + h(-\kappa + \lambda\theta + .00125\lambda^2 - \rho)\}.$$

The choices $(|\lambda|, \kappa) = (120, 42), (160, 64)$ satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore have nonpositive killed generators for every admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) = -1$. It follows that U has nonpositive residual inside the verification domain.

At $t = 1$, $F = Z = C_\rho = 0$, and $U = G$. At an actual wealth exit, $F \geq -8h$ implies $U \geq G - 8h = g$. At either artificial preference face, one summand of Z is at least $8h$, so $U \geq G$. Stop first at the economic exit or an artificial face. If the artificial face occurs first, the remaining original-model payoff is bounded by G under every admissible continuation, by the global fallback just proved. Concatenated stopped verification gives the asserted bound without modifying the original dynamics or opportunity set. *Q.E.D.*

B.2. Polynomial source enclosure and refinement

Write $u - 1 = A + Bz$, where $A = 1.05$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and $q = B/(A + D) < 1$. The uniformly convergent expansion

$$\frac{1}{A + Bz} = \frac{1}{D} \left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z) \right]$$

has truncation error at degree 16 bounded by $2q^{17}/\{D(1 - q)\}$. If $L = -\log(.8)$, approximate $e^{L(A+Bz)}$ by e^{LA} times the degree-eight Taylor polynomial of e^{LBz} . Its remainder is at most $e^{L(A+B)}(LB)^9/9!$. Multiplying the two enclosures, with the relevant sup-norm bounds, gives the stored uniform error below 2.117×10^{-9} for $-.8^{-(u-1)}/(u - 1)$. This error is added to the residual upper bound, not fitted away.

The H_k branches are $-.2q - .02k$ for $q \leq -.2k$, $q^2/(2k)$ for $|q| \leq .2k$, and $.2q - .02k$ for $q \geq .2k$. On each box, interval Bernstein coefficients enclose q and every potentially applicable residual polynomial. A box touching a switching threshold includes both adjoining branches. The comparison uses outward threshold intervals; binary64 representation of $.2k$ cannot exclude a true branch. A tensor polynomial lies between its minimum and maximum Bernstein coefficients. De Casteljau subdivision preserves this enclosure and is repeated on boxes whose upper coefficient exceeds the prescribed ceiling. The independent check of $F/h \geq -8$ uses the same convex-hull property.

For a fixed polynomial whose true residual lies strictly below a ceiling, Bernstein refinement eventually verifies that ceiling, up to the explicitly accounted source and arithmetic allowances. This is a certification statement about a fixed candidate with slack. It is not a theorem that residual fitting, a wealth-independent ansatz, or arbitrary Adam training produces approximants

converging to the optimal value. The executed ceiling sequence and all visited-cell counts are retained. The separate wealth-sensitive linear-programming pilot has not received this validation and contributes no upper endpoint to the reported results.

APPENDIX C: INDEPENDENT EVALUATION OF THE LEARNED TIME POLICY

The projected policy has constant c_j, θ_j on each of sixteen slabs, $p = 0$, and $0 \leq \theta_j \leq .2$. Training chooses these constants, but the subsequent evaluator reads only the frozen policy. The exact wealth formula in the main text and the outward budget check show $x_1 > .5$. Since $c_j > .025$ and $x_0 = 1.25$, wealth is decreasing throughout the horizon and never reaches the upper boundary. Neither Brownian correlation nor a wealth discretization affects this conclusion when $p = 0$.

Without preference stopping, $u_t = 2 + \mu(t) + .05W_t^u$, with $\mu(t) = \int_0^t \theta_r dr \in [0, .2]$. An upper-boundary exit requires $W_t^u \geq 12$ at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is also contained in $\{\sup_{t \leq 1} |W_t^u| \geq 12\}$. The exponential martingale maximal inequality on each side gives

$$p_{\text{exit}} \leq 2e^{-72} < 1.077 \times 10^{-31}.$$

The original preference diffusion is still stopped at exact first exit. Its rare-exit correction is not set to zero.

Put $r = u - 2$ and $L_j = -\log c_j$. Before exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r} / (1 + r)$. Replace the reciprocal by $\sum_{m=0}^{80} (-r)^m$ and the exponential by its degree-ten Taylor polynomial. The geometric remainder is at most $.8^{81}/.2$; the exponential remainder is at most $e^{.8L_j} (.8L_j)^{11}/11!$. The product bound, including the reciprocal and exponential suprema, yields a uniform error for the degree-90 polynomial P_j . This bound holds on the whole original preference interval, not just at training quadrature nodes.

For a Gaussian random variable with mean μ and variance v , its raw moments satisfy $M_0 = 1$, $M_1 = \mu$, and

$$M_n = \mu M_{n-1} + (n-1)v M_{n-2}.$$

Hence $\mathbb{E}P_j(u_t - 2)$ is evaluated by this recurrence with $v = .0025t$, and the terminal quadratic preference settlement has an exact Gaussian expectation. On each time rectangle the intervals for μ , v , discounting, and the polynomial coefficients enclose the entire integrand. Outward sums of interval range times interval length therefore enclose its integral. These ranges are not quadrature-error estimates inferred from two discretizations. The three resolutions are independent enclosures of the same integral.

To compare this free-Gaussian polynomial calculation with the stopped economic payoff, let E be the event of preference exit. Off E the two paths agree through T . On E , the omitted polynomial tail is bounded using

$$\mathbb{E}[\mathbf{1}_E |P_j(u_t - 2)|] \leq \sqrt{p_{\text{exit}}} \|P_j(u_t - 2)\|_2, \quad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

Even Gaussian moments are bounded uniformly by evaluating their nonnegative coefficient formulas at $\mu = .2$ and $v = .0025$. The same argument bounds the free terminal settlement, while actual exit settlement has magnitude below 8.1. Discounted adjustment cost omitted after exit is at most $.02k\mathbf{1}_E$. One valid total stopping allowance is thus

$$\sqrt{p_{\text{exit}}} \left\{ \max_j \|P_j\|_2 + \|G(u_1, x_1)\|_2 \right\} + p_{\text{exit}} (.02k + 8.1).$$

The uniform utility approximation allowance is added separately, multiplied by $C_\rho(1)$. At $k = 2$ the stopping allowance is below 5.388×10^{-16} ; the utility approximation allowance is below 1.151×10^{-7} before multiplying by $C_\rho(1)$. The larger remaining width comes from the explicitly enclosed time integration, not from ignoring a boundary event.

The free adjustment budget is exactly $\sum_j (\theta_j^2/2) \int_{j/16}^{(j+1)/16} e^{-.04t} dt$. Stopping can reduce this quantity by at most $.02p_{\text{exit}}$. Its interval, and the strictly positive wealth-feasibility margin, are retained at full precision. Tight budget intervals describe these deployed policies, not optimal-budget identification.

APPENDIX D: ECONOMIC COMPARISON AND THE RESTRICTED DUAL

D.1. Cost ordering and value enclosures

PROOF OF THEOREM 3: For each policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing because $B(\pi) \geq 0$. The supremum is convex and nonincreasing. The two approximate optimality inequalities give

$$A_1 - k_1 B_1 \geq A_2 - k_1 B_2 - \varepsilon_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2.$$

Adding and dividing by $k_2 - k_1 > 0$ gives (17). Zero allowances give budget ordering for any optimizers at distinct prices. If π_k is optimal at k , feasibility at k_+ implies $V(k_+) \geq V(k) - (k_+ - k)B(\pi_k)$; feasibility at k_- implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute the appropriate value-enclosure endpoints and intersect with the primitive budget range to obtain (18). Q.E.D.

The access lower bound follows from $V_k \geq J_k$ and $V_R \leq \bar{V}_R$. Its nonnegativity follows because the restricted policy set is contained in the flexible policy set. Likewise $V_k - V_R \leq \bar{V}_k - \underline{J}_R$. These inequalities compare two optimal values even though the feasible policies supplying the lower bounds need not be optimal.

D.2. Cancellation of the portfolio and stopped localization

PROOF OF THEOREM 4: For Y as specified, quadratic covariation gives

$$d(Y_t x_t) = (.04Y_t x_t - Y_t c_t)dt + \text{local martingale}.$$

The $.06pYx$ portfolio drift cancels the $-.06pYx$ covariation term for every adaptive p . Including $-.5Y$, the discount term, and the preference quadratic yields

$$R_{c,p}W_0 = \frac{c^{1-u}}{1-u} - yc + .01y - .00005 - .004\log(.5) + .0008(u-2)^2 \leq S(u, y).$$

There is no W_0 mixed $u-x$ or $u-y$ derivative. Correlation is retained in the transition law of (U, Y) used to evaluate the source potential.

For a source lower bound use the feasible consumption .35: its utility on $u \in [1.5, 2.6]$ is bounded below by $-2/\sqrt{.35}$, while $-.35y \geq -4.2$. The remaining terms sum to a nonnegative quantity on this rectangle. Thus $S > -7.581 > -8$. For an upper bound discard $-yc$, maximize utility at $c = .8, u = 2.6$, use $.01y \leq .12$, and bound the remaining terms by .0033. This gives $S < -.7698 < 0$. The clipped extension is bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

The discounted Feynman–Kac potential satisfies the conditional expectation identity for its remaining source integral. Consequently $e^{-\rho r} F_R(h - r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is

a martingale. Stopping this identity on a bounded verification rectangle is legitimate because the source is bounded. Inside the rectangle $\tilde{S} = S$, so adding the identity to discounted Itô verification for W_0 cancels its residual majorant. This argument does not require an assumed globally smooth optimal value, or differentiability of the clipped source at its outer faces.

Preference localization with $\theta = 0$ uses $\kappa = .00125\lambda^2$, giving 50 and 72 for $|\lambda| = 200$ and 240. For $l = \log Y$, $dl = -.025dt - .3dZ^x$. Each exponential localization term with $|\lambda| = 22$ has nonpositive killed generator because $22.33 = .025(22) + .045(22)^2$ dominates either sign of the log drift. As before, the factor $8h$ supplies at least $8h$ at its associated artificial face.

For $x \in [.5, 2]$ and $y \geq .2$,

$$W_0 - G = y(x - .5) + .1 \log(.5/x) \geq 0 :$$

the expression is zero at $.5$ and has derivative $y - .1/x \geq 0$. At an actual wealth exit the source term is at least $-8h$, so $W_0 + F_R + Z_R \geq g$. At a terminal time it is at least G . At any artificial u or y face a localization term pays for the entire possible source deficit, giving continuation value at least G . Actual preference exit cannot precede these inner preference faces. Invoke the global fallback $V_R \leq G$ after an artificial exit and apply the stopped identity before it. This proves (25) over all original adaptive consumption and portfolio controls. The value of Y_0 is an auxiliary witness parameter, not an imposed wealth price or policy restriction. *Q.E.D.*

D.3. Validated evaluation of the restricted witness

Let z_1, z_2 be independent standard Gaussians and put $r = v^2$. At the central state,

$$u = 2 + .05v\{-.25z_1 + \sqrt{.9375}z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

The source expectation is integrated with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box $z_1, z_2 \in [-6, 6]$ lies strictly inside the stated u, y verification rectangle under this transformation. The omitted probability is at most $4e^{-18}$. Since the extended source lies in $[-8, 0]$, omission contributes an interval $[-8C_p(1)4e^{-18}, 0]$, whose width is less than 4.778×10^{-7} . The initial localization addition is less than 2.200×10^{-9} .

For completeness, the Hessian used by the evaluator is explicit. Let $c = \text{proj}_{[.05, .8]} e^{-l/u}$, $L = -\log c$, $y = e^l$, and let χ be one for an interior consumption solution and zero for a strictly constrained solution. Across a switching box use $\chi \in [0, 1]$. With $Q = c^{1-u}/(1-u)$,

$$\begin{aligned} Q_u &= \frac{e^{L(u-1)}\{1 - L(u-1)\}}{(u-1)^2}, & Q_{uu} &= -\frac{e^{L(u-1)}[\{L(u-1) - 1\}^2 + 1]}{(u-1)^3}, \\ S_u &= Q_u + .0016(u-2), & S_l &= -yc + .01y, \\ S_{uu} &= Q_{uu} + \chi yc L^2/u + .0016, & S_{ul} &= -\chi yc L/u, \\ S_{ll} &= -yc + \chi yc/u + .01y. \end{aligned}$$

The envelope is continuously differentiable across clipping surfaces; its gradient is locally Lipschitz and these piecewise second derivatives give a valid integral Taylor bound across such a surface. Differentiate the displayed (u, l) transformation by the chain rule to obtain a Hessian in (v, z_1, z_2) . Every intermediate expression is enclosed on the entire box. Bounds on the absolute Hessian entries give the usual one-half diagonal and full mixed-term Taylor remainder.

The constant and linear Taylor terms are integrated exactly for their polynomial approximation to the weight. Gaussian weighted moments through order two use 111 terms of $e^{-z^2/2}$ and

remainder bounded on $|z| \leq 6$ by $e^{18}18^{11}/11!$. Time moments use thirteen terms of $e^{-.04v^2}$ with their analytic remainder. Source exponentials are evaluated with degree-40 Taylor polynomials on arguments in $[-3, 3]$, with error at most $e^33^{41}/41!$. All antiderivatives, remainders, interval additions, and sums are rounded outward. The two diagonal second moments are non-negative; intersecting their enclosures with the nonnegative half-line is valid. Mixed moments are conservatively bounded by cell-radius products times probability mass.

The successive box counts are 16,384, 131,072, and 1,048,576. Their intervals enclose the same initial *dual witness*. Only an upper endpoint is used to bound the restricted optimum. Combining the final upper endpoint with the independently certified restricted-policy lower gives a true optimal-value interval of width less than .009047. The distinct width below .000320 describes integration of the witness, not the primal–dual gap.

APPENDIX E: CONTINUOUS ACTION AND ABSOLUTE-VALUE CERTIFICATION

E.1. *The scalar global action bound and interval neural jets*

PROOF OF THEOREM 5: For any action with $\sum_i a_i = z$ and any real λ ,

$$\sum_i f_i(a_i) = \sum_i \{f_i(a_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

The second derivative of $.3 \sin z$ is bounded below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic Taylor minorant gives

$$\lambda z + .3 \sin z \geq \lambda z_0 + .3 \sin z_0 + (\lambda + .3 \cos z_0)e - .15e^2,$$

which implies (30). Each scalar f_i has second derivative $(2 + .2 \cos 2s)/d \geq 1.8/d = m$. Thus $q = f_i - \lambda s$ is m -strongly convex. Its tangent quadratic at any feasible s_0 gives a lower bound $q(s_0) - q'(s_0)^2/(2m)$. At an endpoint, only a feasible descent direction is relevant: an outward-pointing minimization direction contributes zero, while an inward-pointing direction contributes the derivative magnitude. This follows either by minimizing the tangent quadratic over the half-line or directly from the sign of the linear term. The stated lower bound remains valid if the quadratic minimizer lies beyond the opposite endpoint, because relaxing the displacement can only lower it.

The sum of these scalar lower bounds is no larger than $D(\lambda)$. A feasible action gives a rigorous upper bound by direct interval evaluation. The scalar interval $[-d, d]$ covers every possible action sum. Subdivision, pruning only a box whose lower bound exceeds a feasible upper bound, and taking the minimum surviving lower bound preserve a global enclosure. Root finding affects efficiency but not validity. Finally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box. The true candidate upper and true minimum lower can each worsen by at most this amount, giving the $4 \sum_i \delta_i$ gap correction. Q.E.D.

For an affine layer with stored weights, interval automatic differentiation propagates both activations and spatial derivatives. A tanh layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a strictly positive denominator and derivative $1 - \tanh^2 z$, all in multiprecision intervals. The terminal embedding is differentiated as its mathematical formula, including the exact $\sqrt{d}$, not as a previously rounded autograd constant. This gives a componentwise interval for the derivative of the represented network. Subtracting the stored floating-point gradient gives the δ_i . The two executed tests use the inherited eight- and sixteen-dimensional learned checkpoints and the same jets as the action certificates. The summed errors are below 4.584×10^{-17} and 7.877×10^{-17} respectively. These are local jet checks, not whole-state residual bounds.

E.2. A lower bound retaining the nonquadratic terminal payoff

PROOF OF PROPOSITION 3: For $|a| \leq 1$, $\sin(a)/a \geq \sin 1$, with the continuous value at zero. Consequently $a^2 + .1 \sin^2 a \geq r_0 a^2$, where $r_0 = 1 + .1 \sin^2 1$. Also $.3 \sin(\sum_i a_i) \geq -.3$. Replacing running cost by $r_0 \|a\|^2/d - .3$ decreases every original control's cost; allowing all $a \in \mathbb{R}^d$ decreases the infimum further. The original terminal payoff q_d is unchanged.

The relaxed HJB equation is

$$v_t + \frac{\sigma^2}{2} \Delta v - \frac{d}{r_0} |\nabla v|^2 - .3 = 0, \quad v(1, x) = q_d(x), \quad \sigma^2 = .32.$$

Put $\lambda = \sigma^2 r_0 / (2d)$ and use the logarithmic heat representation. At $X_0 = 0$ it gives

$$v(0, 0) = -.3 - \lambda \log \mathbb{E} \exp\{-q_d(\sigma Z)/\lambda\}, \quad Z \sim N(0, I_d).$$

The bounded cosine and coercive quadratic terminal terms give finite positive expectations. The Gaussian heat solution is smooth before the terminal date and has at most quadratic growth. Standard localization of its verification identity, or truncation followed by monotone Gaussian integration, validates the relaxed lower bound without assuming a smooth solution of the original bounded-control problem.

Complete the square for the quadratic terminal component. Its Gaussian normalization contributes

$$\frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

Under the tilted Gaussian law the scalar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and variance $s^2 = \sigma^2 r_0 / (r_0 + 4)$. Orthogonal coordinates disappear, leaving

$$v(0, 0) = \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

This is the representation in the proposition. An upper enclosure for the positive scalar expectation gives a lower enclosure for v , because $-\lambda \log$ is decreasing. That lower endpoint is therefore a lower bound for the original value. Q.E.D.

For the scalar integral let $\eta = .2/\lambda$ and $R = \lceil \sqrt{2(\eta + 50)} \rceil$. The omitted Gaussian contribution is no larger than $2e^{\eta - R^2/2}$. Interval ranges of the positive integrand times cell lengths enclose the integral on $[-R, R]$. The successive 1,024, 4,096, and 16,384-cell calculations add this analytic tail allowance and then apply the decreasing logarithm with outward rounding. Their widths measure integration uncertainty in the relaxed value. Relaxing the action bound and the running sine can leave a large structural gap even when this quadrature width is small.

APPENDIX F: REPLICATION AND SCOPE LEDGER

The central economic certificate is independent of the inherited killed-Euler discretization. Its fitted upper has a verified continuous-generator residual, an explicit source-polynomial remainder, a stopped-trace inequality, and exponentially bounded artificial-face continuation. Its feasible lower has exact deterministic wealth, interval Gaussian polynomial moments, an explicit utility approximation remainder, outward time integration, and an exact-exit probability correction. The restricted upper cancels continuous portfolio controls algebraically and evaluates

a bounded-source stochastic potential with Taylor, Gaussian-tail, and localization allowances. These routes do not establish the still-uncomputed operator discrepancy budget for the old finite scheme.

The external experiment has a different status. Each declared sample-and-hold policy is a feasible continuous control in the stated additive-noise dynamics. Real-arithmetic transitions are exact per holding slab, but the stored path estimates use floating-point network evaluations and a finite Monte Carlo sample. The rigorous lower bound supplies an absolute reference; subtracting it from a sampled mean does not create an outward policy-regret certificate. Paired path errors, seed variation, completed-update clock overshoot, setup, and warmup are recorded separately. The exact sign test concerns the frozen joint training-and-audit pipeline under independent seed draws. It does not certify an underlying PDE or imply that the six panels represent six independent problem families.

Every new table is generated from raw JSON or path records; display endpoints are rounded in the conservative direction. The frozen external protocol, all tuning trials, all 72 holdout pairs, all checkpoints, and all path arrays are retained. The validation program checks coverage of the declared panels, matching initialization hashes, pinned author revision, finite outputs, refinement ordering, primitive feasibility, the positive access lower bounds, and the explicitly false flexible .01 precision flags. The complete R8, R4, and delivered R5 textual and numerical material is included in the preservation supplement, and earlier source files remain byte-for-byte unchanged. Neither a failed pilot nor a historical manufactured-payoff test is used as new original-economy evidence.

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